

Open-system model for light-shift-driven multiparameter spin metrology

1 Setup

We consider a single spin J (one atom of a product ensemble of N_a atoms; for $J = 3$ this is the Cs $F = 3$ hyperfine manifold) evolving under a closed-system Hamiltonian that is the sum of a static field to be estimated, an rf control rotation with a piecewise-constant random phase, and a nonlinear term generated by the tensor light shift of an off-resonant, linearly polarized beam. With $\hbar = 1$ and time in units of $1/\Omega$,

$$H(t) = \underbrace{\mathbf{w} \cdot \mathbf{J}}_{\text{field to estimate}} + \underbrace{\Omega [\cos \phi(t) J_x + \sin \phi(t) J_y]}_{\text{rf control}} + \underbrace{\frac{\beta}{2J} J_z^2}_{\text{tensor light shift}}, \quad (1)$$

where $\mathbf{w} = (w_x, w_y, w_z)$ is the three-component parameter vector and $\phi(t)$ is constant on blocks of `random_phase_steps` time steps and drawn uniformly on $[0, 2\pi)$ for every realization. The code's parameter `beta` enters as $\beta/2J$, as written.

2 Light-induced decoherence

The same beam that produces the J_z^2 term scatters photons. In the far-detuned limit (detuning large compared with the excited-state hyperfine splitting Δ'_{HF} but small compared with the fine-structure splitting) the scattering amplitude acts on the electron spin alone. For linear polarization the no-spin-flip (Rayleigh) part is a scalar and does not decohere the ground manifold. The spin-flip (Raman) part, projected onto the F manifold, is proportional to $J_{\pm}$ by the Wigner–Eckart theorem. Neglecting Raman loss to the other hyperfine manifold and any decoherence from a separate readout beam, the single decoherence channel is therefore the pair of jump operators

$$L_+ = \sqrt{\gamma} J_+, \quad L_- = \sqrt{\gamma} J_-, \quad J_{\pm} = J_x \pm iJ_y. \quad (2)$$

Both the tensor light shift and the scattering rate scale with intensity over detuning squared, $\beta \propto I \Delta'_{\text{HF}}/\Delta^2$ and $\gamma_s \propto I \Gamma/\Delta^2$, so their ratio is a property of the atom and the transition, not of the beam. We parametrize this by a fixed figure of merit

$$\eta \equiv \frac{\beta}{\gamma}, \quad \gamma = \frac{\beta}{\eta}, \quad \eta \sim c \frac{\Delta'_{\text{HF}}}{\Gamma} = \mathcal{O}(10\text{--}100), \quad (3)$$

with c an order-one Clebsch–Gordan factor obtainable from the ground-state polarizability and optical-pumping formalism of Deutsch and Jessen (Opt. Commun. **283**, 681 (2010)). Consequently the nonlinearity cannot be increased without increasing the decoherence in proportion, and a sweep over β is simultaneously a sweep over γ .

3 Master equation (Schrödinger picture)

The Lindblad master equation for the single-atom density matrix is

$$\frac{d\rho}{dt} = -i[H(t), \rho] + \gamma \left(J_+ \rho J_- - \frac{1}{2} \{J_- J_+, \rho\} \right) + \gamma \left(J_- \rho J_+ - \frac{1}{2} \{J_+ J_-, \rho\} \right). \quad (4)$$

Writing $J_{\pm} = J_x \pm iJ_y$, the cross terms cancel and the dissipator collapses to

$$\mathcal{D}[\rho] = 2\gamma \sum_{k=x,y} \left(J_k \rho J_k - \frac{1}{2} \{J_k^2, \rho\} \right) = -\gamma \left([J_x, [J_x, \rho]] + [J_y, [J_y, \rho]] \right), \quad (5)$$

a double-commutator (dephasing-type) channel along x and y , each at rate 2γ . It commutes with rotations about z , conserves the trace, and is unital: the maximally mixed state $\mathbb{K}/(2J+1)$ is its fixed point. Here $J_- J_+ + J_+ J_- = 2(J_x^2 + J_y^2) = 2[J(J+1) - J_z^2]$.

4 Master equation (Heisenberg picture)

Expectation values can equally be computed by propagating the observable with the adjoint generator,

$$\langle O \rangle(t) = \text{Tr}[O e^{t\mathcal{L}}(\rho_0)] = \text{Tr}[e^{t\mathcal{L}^\dagger}(O) \rho_0], \quad (6)$$

with

$$\begin{aligned} \frac{dO}{dt} &= i[H(t), O] + \gamma \left(J_- O J_+ - \frac{1}{2} \{J_- J_+, O\} \right) + \gamma \left(J_+ O J_- - \frac{1}{2} \{J_+ J_-, O\} \right) \\ &= i[H(t), O] - \gamma \left([J_x, [J_x, O]] + [J_y, [J_y, O]] \right). \end{aligned} \quad (7)$$

The dissipator takes the same form in both pictures because the jump set $\{J_+, J_-\}$ is closed under Hermitian conjugation. This is the form integrated in the simulation: the observable $O = J_y$ (or J_z) is propagated and the signal is $\text{Tr}[O(t)\rho_0]$ with ρ_0 the initial spin coherent state.

Exact relaxation of the first moments

Because $\mathcal{D}^\dagger$ maps the vector operator $\mathbf{J}$ onto itself,

$$\mathcal{D}^\dagger[J_x] = -\gamma J_x, \quad \mathcal{D}^\dagger[J_y] = -\gamma J_y, \quad \mathcal{D}^\dagger[J_z] = -2\gamma J_z, \quad (8)$$

so for any J the Bloch vector obeys

$$\frac{d}{dt} \langle J_x \rangle = (\text{coherent}) - \gamma \langle J_x \rangle, \quad (9)$$

$$\frac{d}{dt} \langle J_y \rangle = (\text{coherent}) - \gamma \langle J_y \rangle, \quad (10)$$

$$\frac{d}{dt} \langle J_z \rangle = (\text{coherent}) - 2\gamma \langle J_z \rangle. \quad (11)$$

The coherent part is not closed at first order because the J_z^2 term couples $\langle \mathbf{J} \rangle$ to second moments; it is handled numerically below. The parameter γ is therefore the transverse relaxation rate, and 2γ the longitudinal one.

5 Measurement model

At each time step n the recorded datum is

$$M_n = \langle O \rangle_n + \xi_n, \quad \xi_n \sim \mathcal{N}(0, \sigma^2), \quad \sigma = \frac{1}{\sqrt{N_p} G N_a}, \quad (12)$$

i.e. the ensemble-averaged single-atom expectation value with Gaussian photon shot noise from a separate readout beam of N_p photons per step, Faraday gain G , and N_a atoms. Back-action of the readout beam is not included. The estimator is the least-squares fit of $\mathbf{w}$ to $\{M_n\}$ using the same open-system forward model, with γ treated as known (it is fixed by the chosen β), and its covariance is compared with the classical Cramér–Rao bound $\text{Cov}(\hat{\mathbf{w}}) \geq F^{-1}$,

$$F_{ab} = \frac{1}{\sigma^2} \sum_n \frac{\partial \langle O \rangle_n}{\partial w_a} \frac{\partial \langle O \rangle_n}{\partial w_b}. \quad (13)$$

6 Numerical integration

Time is discretized with step δt . With $U_w = e^{-i\delta t \mathbf{w} \cdot \mathbf{J}}$ and $U_c[n] = e^{-i\delta t H_c[n]}$ the control step for block n , define $U_n = U_c[n] U_w$. The Hamiltonian and dissipative parts are split to first order,

$$\text{Schrödinger:} \quad \rho_{n+1} = e^{\delta t \mathcal{D}} (U_n \rho_n U_n^\dagger), \quad (14)$$

$$\text{Heisenberg:} \quad O_{n+1} = U_n^\dagger e^{\delta t \mathcal{D}^\dagger} (O_n) U_n. \quad (15)$$

The map $A \equiv e^{\delta t \mathcal{D}^\dagger}$ is built once per realization as a $(2J+1)^2 \times (2J+1)^2$ matrix acting on the vectorized observable (the exact solution of the dissipative part over one step), so no approximation beyond the first-order splitting is made. That splitting is of the same order as the pre-existing split between U_w and U_c . Because A is linear and independent of $\mathbf{w}$, the Fréchet-derivative recursion for $\partial O_n / \partial w_a$, needed for the Fisher matrix, is unchanged apart from passing O_n and $\partial O_n / \partial w_a$ through A first. Setting $\gamma = 0$ (no `--eta`) recovers the closed-system code exactly.

Expected consequence. The signal derivatives $\partial \langle O \rangle_n / \partial w_a$ decay on the time scale $1/\gamma$, so the accumulated Fisher information saturates near $T \sim 1/\gamma$ instead of growing polynomially. At a fixed total time T the useful nonlinearity is bounded by $\beta \lesssim \eta/T$, and the question becomes whether the nonlinearity improves multiparameter estimation enough to pay for the decoherence it necessarily brings.

7 Assumptions and how to relax them

The model above rests on two simplifications, adopted deliberately for the first study. Both can be removed within the same numerical scheme.

A1: the readout beam does not decohere the atoms

Relaxed in Section 8 (script v6).

The measurement model treats the N_p readout photons per step as pure shot noise. A readout beam scatters photons too, through the same Raman channel (2), so it adds to γ a rate γ_r set by the readout intensity rather than by β . Because Faraday signal and scattering both scale as

$1/\Delta^2$, the ratio of measurement signal-to-noise to scattering is independent of detuning and fixed by the resonant optical depth $\text{OD} = N_a \sigma_0 / A$: up to order-one polarization and branching factors, the number of photons scattered per atom over the whole run is

$$P_{\text{sc}} \simeq \frac{\text{SNR}_{\text{tot}}^2}{N_a \text{OD}}, \quad \text{SNR}_{\text{tot}}^2 = G^2 N_a^2 N_p^{\text{tot}} = \sum_n \sigma^{-2} \quad (16)$$

per unit of $\langle J \rangle$. With the values used in the simulations ($G = 8.9 \times 10^{-7}$, $N_a = 2 \times 10^5$, $N_p^{\text{tot}} = 9.6 \times 10^8$) this gives $\text{SNR}_{\text{tot}}^2 \approx 3 \times 10^7$ and hence $P_{\text{sc}} \sim 10^2 / \text{OD}$, i.e. $\gamma_r T \sim 10^2 / \text{OD}$. For comparison the light-shift beam contributes $\gamma T = \beta T / \eta$, which for $T = 14 / \Omega$ and $\eta \sim 30$ is $\approx 0.5 \beta$. So unless $\text{OD} \gtrsim 10^3$ (a cavity, or a very dense cloud), the readout beam decoheres the atoms at least as strongly as the light-shift beam does for $\beta \lesssim 1$, and assumption A1 is not self-consistent with the assumed shot-noise level. Relaxing it costs one extra parameter, OD, and one change to the model: replace $\gamma \rightarrow \gamma + \gamma_r$ in (2) with $\gamma_r = P_{\text{sc}} / T$, independent of β . This also introduces the usual continuous-measurement trade-off between N_p and γ_r that the present model does not see. (In the single-beam scheme of Deutsch and Jessen the light-shift and readout beams are the same beam, so β , N_p and γ are all tied to one intensity.)

A2: no loss from the F manifold

Relaxed in Section 9 (script v7).

The Raman process that gives (2) also transfers atoms to the other ground hyperfine manifold ($F = 4$ for Cs $F = 3$) with a branching ratio of order one, so loss occurs at a rate $\gamma_L \sim \gamma$. Within the F -manifold Hilbert space this is a non-trace-preserving term

$$\frac{d\rho}{dt} \supset -\frac{1}{2} \{K_L, \rho\}, \quad K_L = \sum_q L_q^{L\dagger} L_q^L = a + c J_z^2, \quad (17)$$

where $L_q^L \propto P_{F'} s_q P_F$ are the jump operators into F' and the form of K_L follows from z -symmetry and the $m \rightarrow -m$ reflection symmetry of π -polarized light. If the lost atoms are dark to the readout, the unnormalized $\text{Tr}[O\rho]$ is automatically the measured signal, and in the simplest uniform approximation $K_L = \gamma_L \mathbb{K}$ this is just a known factor $e^{-\gamma_L t}$ multiplying the signal and its derivatives. In the code this is one additional w -independent factor per step, so the fit and Fisher recursion are again unchanged. The complication that this approximation hides is that atoms in F' are not dark: they Faraday-rotate the readout with a different (roughly opposite) sign and evolve under a different light shift. Treating that properly requires enlarging the state space to $F \oplus F'$ (dimension 16 for Cs), which the superoperator scheme of Section 6 supports at the cost of a 256×256 dissipative map.

8 One beam: light shift, decoherence and readout from the same probe

In the actual scheme there is a single off-resonant, linearly polarized probe. It produces the tensor shift, it scatters photons, and its Faraday rotation is the readout. With photon flux Φ and detuning Δ , in the far-detuned limit,

$$\beta \propto \frac{\Phi \Delta'_{\text{HF}}}{\Delta^2}, \quad \gamma \propto \frac{\Phi \Gamma}{\Delta^2}, \quad G \propto \frac{1}{\Delta}, \quad \sigma^2 = \frac{1}{N_p G^2 N_a^2} \propto \frac{\Delta^2}{\Phi}, \quad (18)$$

where G is the rotation per atom per unit spin and $N_p \propto \Phi$ the photons per measurement interval δt . All four depend on Φ and Δ only through Φ/Δ^2 , so the probe strength is a single knob and two ratios are fixed by the physics:

$$\frac{\beta}{\gamma} = \eta, \quad \gamma \sigma^2 \delta t = \frac{1}{\text{OD}_{\text{eff}} N_a}, \quad \text{OD}_{\text{eff}} = c \frac{N_a \sigma_0}{A}, \quad (19)$$

with c an order-one factor from the vector polarizability and the spin-flip branching ratio. The second relation is the readout back-action of Section 7 made exact: the photons that carry the Faraday signal are the photons that scatter. Taking β as the sweep variable, the script `v6` derives per β

$$\gamma = \frac{\beta}{\eta}, \quad \sigma(\beta) = \sqrt{\frac{\eta}{\text{OD}_{\text{eff}} N_a \delta t \beta}}. \quad (20)$$

A stronger probe brings more nonlinearity, more decoherence and *less* shot noise; $\beta \rightarrow 0$ is no light at all, with no decoherence but no measurement either, so $\beta = 0$ is excluded from the sweep and there is no decoherence-free reference. The former constants G and N_p are absorbed into OD_{eff} ; with the values used in `v4/v5` and $\eta = 30$, $\text{OD}_{\text{eff}} = 10$, the old noise level $\sigma = 0.021$ corresponds to $\beta \approx 33$, a far stronger probe than any β in the `v5` sweep. The two physics inputs are now η (atom) and OD_{eff} (sample); both are placeholders in the launcher until derived and measured.

Expected shape of the optimum. The Fisher information per step is $(\partial \langle O \rangle / \partial w)^2 / \sigma^2 \propto \beta (\partial \langle O \rangle / \partial w)^2$, and the derivative decays on the scale η/β . For $T \ll \eta/\beta$ the information grows like βT^3 (more probe is better), for $T \gg \eta/\beta$ it saturates like η^3/β^2 (less probe is better), so the optimum sits at a finite $\beta_{\text{opt}} \sim \eta/T$ for every T . Unlike `v5`, where $\beta = 0$ eventually won because it was decoherence-free, the one-beam model always has a nonzero optimal probe strength.

Control experiment. Because the nonlinearity cannot be switched off physically without switching off the probe, `v6` has a flag `--tensor-scale` that multiplies the J_z^2 term in the Hamiltonian only, keeping the beam-derived γ and σ . Running the same sweep with `--tensor-scale 0` isolates what the nonlinear dynamics contributes to the estimation at fixed decoherence and fixed shot noise.

9 Loss, and the light-induced coefficients from atomic structure

The coefficients that Sections 2 and 8 left as placeholders (η , the order-one factors in OD_{eff}) follow from the atomic structure. The script `cs_light_shift_scattering_ratios.py` builds, for Cs ($I = 7/2$) on the D1 or D2 line and a π -polarized far-detuned probe, the ground-state light-shift operator and the optical-pumping jump operators by adiabatic elimination of the excited manifold (Steck matrix elements, closure and normalization checked, vector-to-scalar polarizability ratio converging to $\pm 1/4$ on D1 and $\pm 1/8$ on D2). Writing γ_s for the total photon scattering rate of the spin coherent state, the far-detuned values are

line, F	$\eta_s = \beta/\gamma_s$	r_{within}	$K_L/\gamma_s = a_L + c_L m^2$	b/a	$\eta_{\text{tot}} = \beta/R_{\text{tot}}$
D2, $F = 3$	7.4	0.0055	$0.204 + 0.0107 m^2$	0.127	33
D2, $F = 4$	-9.5	0.0052	$0.130 + 0.0104 m^2$	-0.123	-61
D1, $F = 3$	-100	0.0110	$0.412 + 0.0264 m^2$	-0.243	-217
D1, $F = 4$	122	0.0115	$0.254 + 0.0170 m^2$	0.257	408

Here $r_{\text{within}}\gamma_s$ is the rate of the within-manifold spin-flip channel (2), K_L the rate operator of Raman transfer out of the manifold, b/a the vector-to-scalar polarizability ratio per unit m that sets the Faraday signal, and $R_{\text{tot}} = r_{\text{within}}\gamma_s + \langle K_L \rangle$ the initial relaxation rate of $\langle F_x \rangle$. A negative η_s is a tensor shift of the opposite sign. Two things stand out: the relaxation is 96–98 % loss and only a few per cent spin flips, so the channel of Sections 2–5 is the minor one; and $\eta_{\text{tot}} = 33$ for D2, $F = 3$ is essentially the placeholder $\eta = 30$ used so far, while D1, $F = 4$ offers twelve times more nonlinearity per unit of relaxation.

Master equation with loss. The loss adds a trace-decreasing term with no jump operator, since the atom leaves the manifold:

$$\frac{d\rho}{dt} = -i[H, \rho] + \mathcal{D}_{\text{sf}}[\rho] - \frac{1}{2}\{K_L, \rho\}, \quad \frac{dO}{dt} = i[H, O] + \mathcal{D}_{\text{sf}}^\dagger[O] - \frac{1}{2}\{K_L, O\}, \quad K_L = \gamma_s(a_L + c_L J_z^2), \quad (21)$$

with $\mathcal{D}_{\text{sf}}$ the channel (5) at rate $r_{\text{within}}\gamma_s$. The step map $A = e^{\delta t \mathcal{D}^\dagger}$ of Section 6 absorbs both terms and stays $\mathbf{w}$ -independent; on the identity it acts as $e^{-\delta t K_L}$, so $\text{Tr}[O(t)\rho_0]$ is the survival probability times the conditional expectation, which is the collective Faraday signal if lost atoms are dark. Survivors evolve coherently: loss only scales the signal, unlike the spin-flip channel that scrambles it. The remaining approximation of the light model is that atoms transferred to F' are not in fact dark; they carry a vector response of opposite sign and a different light shift.

Parametrization (script v7). With the probe strength swept as β ,

$$\gamma_s = \frac{\beta}{|\eta_s|}, \quad \gamma_{\text{sf}} = r_{\text{within}}\gamma_s, \quad K_L = \gamma_s(a_L + c_L J_z^2), \quad \sigma^2 = \frac{1}{(b/a)^2 \text{OD}_{\text{res}} N_a \delta t \gamma_s}, \quad (22)$$

where $\text{OD}_{\text{res}} = N_a \sigma_{\text{res}}/A$ is the resonant optical depth of the probe transition, the only sample input besides N_a . In terms of Section 8, $\text{OD}_{\text{eff}} = (b/a)^2 \text{OD}_{\text{res}}/(R_{\text{tot}}/\gamma_s)$, which is 0.07 (D2, $F = 3$) to 0.22 (D1, $F = 4$) of OD_{res} . The $\text{OD}_{\text{eff}} = 10$ of the first one-beam run therefore corresponds to $\text{OD}_{\text{res}} \approx 140$; a free-space cloud of 2×10^5 atoms in a $50 \mu\text{m}$ beam has OD_{res} of order 6–9, so both values are worth running.

Cramér–Rao bookkeeping. The simulation fits every realization on its own control sequence. The bound that such an estimator can reach, averaged over realizations, is $\mathbb{E}[F_i^{-1}]$, not $[\mathbb{E}F_i]^{-1}$ (Jensen); with strong decoherence the information comes from a short window in which the particular control phases matter, and the two differ by an order of magnitude (a factor 17 at $\beta = 20$, $\eta = 30$). Scripts v6 and v7 save both.